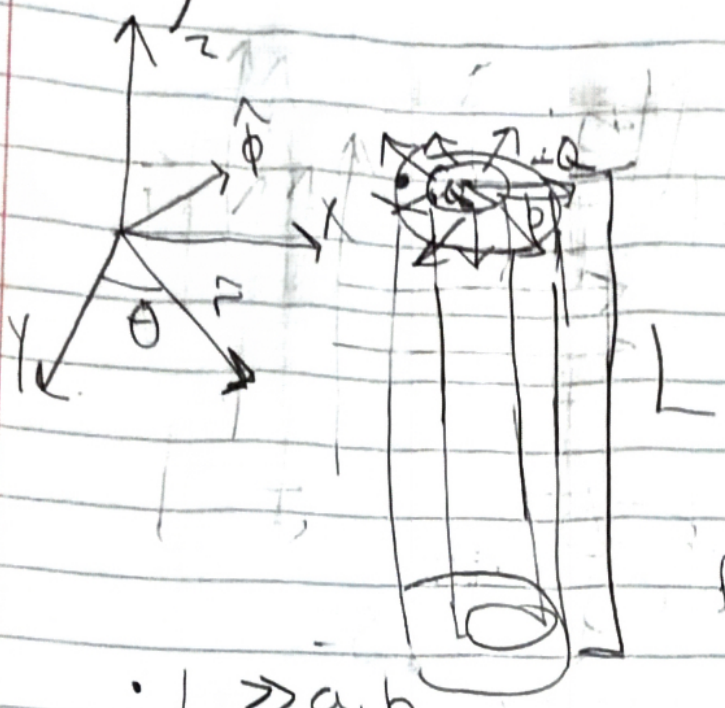
Homework #2

Prob 1.)



objective:

- Determine the capacitance?

C, r, θ, z)

14 words

Rose Raiders 33230116
Rose 33230116!

- $L \gg a, b$

$$C = \frac{Q}{\Delta V}, \quad \cdot \int_S d\vec{s} \cdot \vec{E}(\vec{r}) = \frac{Q_V}{\varepsilon_0}$$

- $A_{cylinder} = 2\pi r(r+l) = 2\pi r l$
- $l \gg r$ (either radius)
- $\vec{E}(\vec{r}) = E_r \hat{r}$
- $d\vec{s} = r\,d\theta\,dz\,\hat{r}$

$0 \leq z \leq L$, $0 < \theta \leq 2\pi$

- $E(\vec{r}) = -\nabla V(\vec{r})$
- $V(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}_1} d\vec{l} \cdot \vec{E}$

- $d\vec{s} = r\,d\theta\,dz\,\hat{r}$
- $\vec{E} = E_r\,\hat{r}$
- $\left(d\vec{s}\cdot\vec{E}\right) = E_r\,r\,d\theta\,dz$
- $E_r r \int_0^{l}\int_0^{2\pi} d\theta\,dz = \frac{Q_{enc}}{\varepsilon_0}$
- $E\,r\,l\,2\pi = \frac{Q_{enc}}{\varepsilon_0}$
- $\frac{Q_{enc}}{2\pi r l \varepsilon_0} = E_r$
- $\vec{E} = \frac{Q_{enc}}{2\pi r l \varepsilon_0}\,\hat{r}$

(cylindrical coordinates)

- $d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + dz\,\hat{k}$
- $\left(d\vec{l}\cdot\vec{E}\right) = \frac{Q_{enc}}{2\pi l \varepsilon_0}\,\frac{dr}{r}$

- $b > a$

- $V(\vec{r}) = -\int_b^a \frac{Q_{enc}}{2\pi l \varepsilon_0} \frac{dr}{r}$

Potential difference is Inner - outer.

- $V(\vec{r}) = \int_a^b \frac{Q_{enc}}{2\pi l \varepsilon_0} \frac{dr}{r} = \frac{Q_{enc}}{2\pi \varepsilon_0 l} \ln(r)\Big|_a^b$

- $V(\vec{r}) = \frac{Q_{enc}}{2\pi \varepsilon_0 l} \ln\left(\frac{b}{a}\right)$

- $C = \frac{Q}{V}$

- $\frac{Q_{enc}}{V} = \frac{2\pi \varepsilon_0 L}{\ln\left(\frac{b}{a}\right)} = C$

Prob 2.)

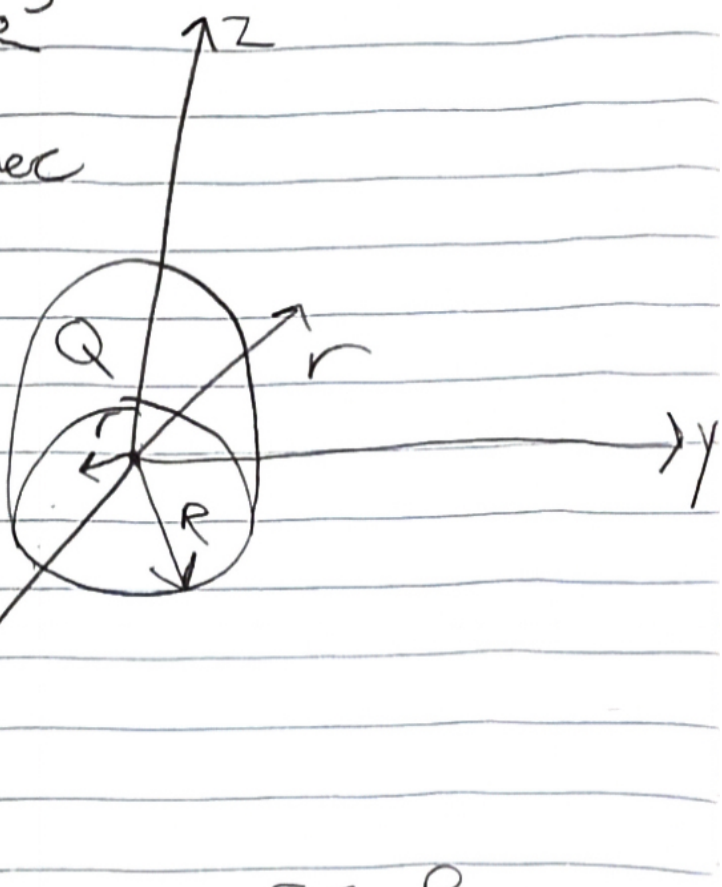
a.)

Givens:

- $\rho = \frac{\text{Total charge}}{\text{volume}}$
- $V_{sphere} = \frac{4}{3}\pi R^3$
- $R \rightarrow$ radius of sphere

objective:

- find $\Phi(r)$ in regions: $0 \leq r < \infty$, $r = 0$, & $r \rightarrow \infty$



$$\rho(r) = \frac{Q}{\left(\frac{4}{3}\pi R^3\right)}$$

$r \leq R$

contribution:

- $\Phi(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} d\vec{l} \cdot \vec{E}$

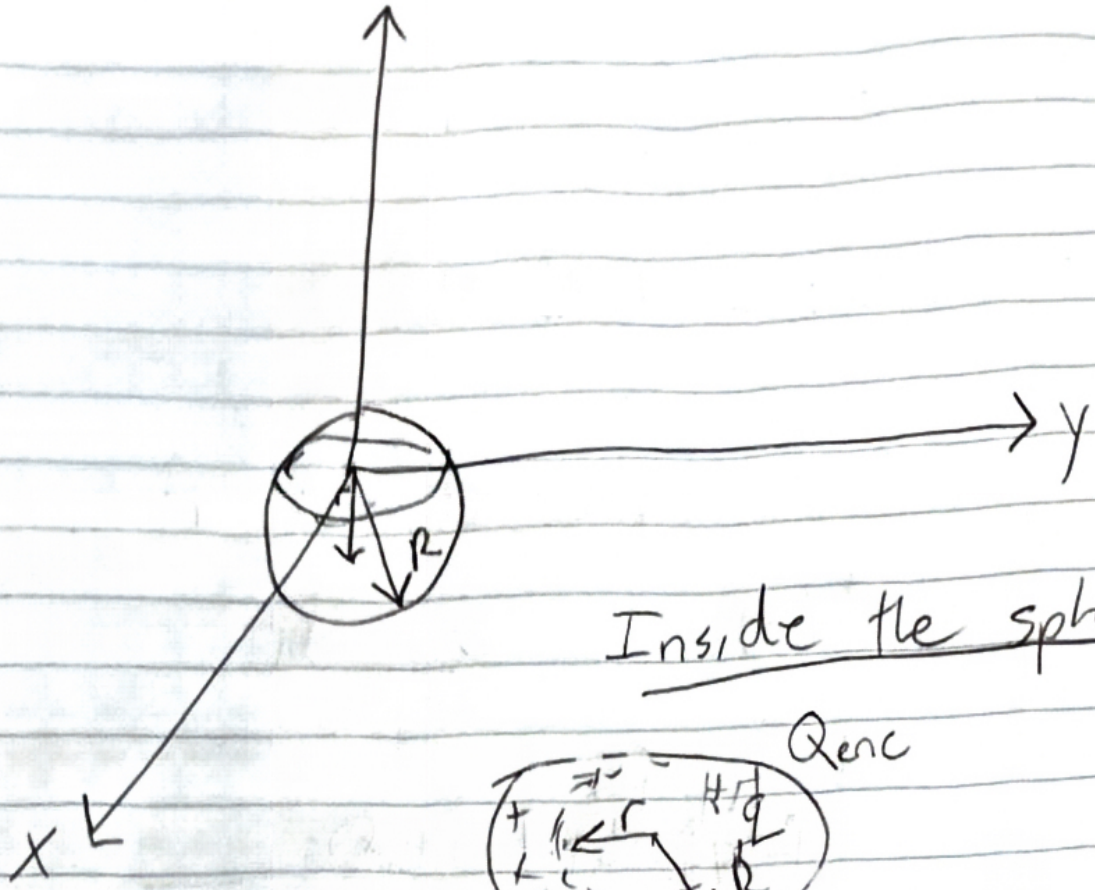
- $\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int_V \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|}\, d^3r'$
  - Contributions inside sphere.

Questions to attempt problem:

1.) Is charge inside sphere, outside, or both?

2.) $0 \le r < \infty$ region is this on infinite sphere, or do I need to evaluate potential difference inside and out?

3.) $r = 0$, do I plug in charge density to my point charge problem, or could it be I plug in $\frac{2}{\infty}$ to general formula? limit



Inside the sphe

Outside the sphere:



Inside Region:

- $\rho_{inside} = \frac{3Q_{enc}}{4\pi R^3}$, $\vec{E} = E_r \hat{r}$
- $\int_A \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$, $E_r 4\pi r^2 = \frac{CQ_{enc}}{\varepsilon_0}$

- $\vec{E} = \frac{Q}{4\pi r^2 \varepsilon_0} \hat{r}$
- $\phi(r) = -\int_A^B \vec{E} \cdot d\vec{l}$

  $\therefore$ $d\vec{l} = dr\,\hat{r}$

- $\phi(r) = -\int_0^r \frac{Q}{4\pi r^2 \varepsilon_0} dr$ $\quad 0 \le r' \le r$
- $\phi(r) = \frac{CQ}{4\pi r \varepsilon_0}$

- $Q \rightarrow$ smaller charge of uniform sphere ($r \le R$ region).
- Relate Q and Q_{enc} (actual sphere)

$\rho_{inside} = \frac{Q}{\frac{4\pi r^3}{3}} = \frac{Q_{enc}}{\frac{4\pi R^3}{3}}$, $Q = \frac{Q_{enc} r^3}{R^3}$

$\vec{E}_{in} = \frac{Q_{enc} r}{4\pi R^3 \varepsilon_0} \hat{r}$

- $\phi(r) = -\int_A^B \vec{E}_{in} \cdot d\vec{l}$

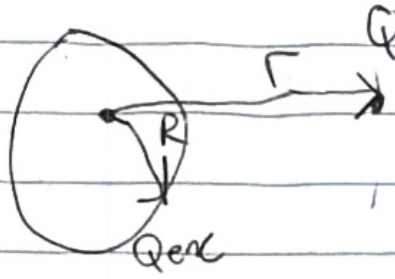
- $\phi(r) = -\int^0 \frac{Q_{enc}\, r}{4\pi\varepsilon_0 R^3}\, dr$

- $\phi(r) = -\frac{Q_{enc} r^2}{8\pi\varepsilon_0 R^3}\Big|_r^0$

$= 0 + \frac{Q_{enc} r^2}{8\pi\varepsilon_0 R^3} + C$

Outside Region: $(r \geq R)$

- Integrate outside to in.



- Q is the same for Qenc because of spherical symmetry

$Q = Q_{enc}$

$E_{outside} = \frac{Q_{enc}}{4\pi r^2 \varepsilon_0}$

- For the outside region for an infinite enclosed sphere of actual you'll still get a point charge potential.

(Large or small region)

- You still get similar point charge potential for $r \geq R$ region and $r \leq R$, if

$$\int_{\infty}^{r} \frac{Q_{enc}}{4\pi\varepsilon_0 r^2}\, dr = \frac{-Q}{4\pi\varepsilon_0 r}$$

$$\frac{-Q}{4\pi\varepsilon_0}\frac{1}{r}\Big|_{\infty}^{r} = \frac{-Q}{4\pi\varepsilon_0 r}$$

$$\phi(r) = \begin{cases} \frac{Q_{enc} r^2}{8\pi\varepsilon_0 R^3} + \frac{3Q_{enc}}{8\pi\varepsilon_0 R}, & r \leq R \\ \frac{-Q_{enc}}{4\pi\varepsilon_0 r}, & r \geq R \end{cases}$$

$r \to \infty$

$$\phi(r) = \frac{Q_{enc}}{4\pi\epsilon_0 r} = 0$$

$r \geq R$

* $r = 0$ region:

- $\phi(R) = |\phi(R)| + |\phi(R)|$
  outside $(r > R)$ inside $(r < R)$
- $\phi(R) = \frac{Q_{enc}}{4\pi\epsilon_0 R} + \frac{Q_{enc}}{8\pi\epsilon_0 R}$
- $$\phi(R) = \frac{3 Q_{enc}}{8\pi\epsilon_0 R}$$

b) c.)

- W = Total region energy inside + Total region energy outside

- $W = \frac{1}{2}\int_V \rho(\vec{r})\,\phi(\vec{r})\,dV$

- $W = \frac{\varepsilon_0}{2}\int_V |\vec{E}|^2\,dV$

- $\vec{E}_{in} = \frac{Q_{enc}\,r}{4\pi R^3 \varepsilon_0}\,\hat{r}$
- $\vec{E}_{out} = \frac{Q_{enc}}{4\pi r^2 \varepsilon_0}\,\hat{r}$

$$|\vec{E}_{in}| = \vec{E}_{in}\cdot\vec{E}_{in} = \frac{Q_{enc}^2\, r^2}{16\pi^2 R^6 \varepsilon_0^2}$$

$$|\vec{E}_{out}| = \vec{E}_{out}\cdot\vec{E}_{out} = \frac{Q_{enc}^2}{16\pi^2 r^4 \varepsilon_0^2}$$

$$W = \underbrace{\int_R^\infty \frac{Q_{enc}^2}{16\pi^2 r^4 \varepsilon_0^2}\left(\frac{\varepsilon_0}{2}\right)(4\pi)\, r^2 dr}_{W_1} + \underbrace{\int_0^R \frac{Q_{enc}\, r^2}{16\pi^2 R^6 \varepsilon_0^2}\left(\frac{\varepsilon_0}{2}\right)(4\pi)\, r^2 dr}_{W_2}$$

$$W_1 = \int_R^\infty \frac{Q_{enc}^2}{8\pi r^2 \varepsilon_0}\, dr$$

$r^{-2} \rightarrow -r^{-1}$

$$W_1 = \frac{-Q_{enc}^2}{8\pi\varepsilon_0}\frac{1}{r}\Big|_R^\infty = 0 + \frac{Q_{enc}^2}{8\pi\varepsilon_0 R}$$

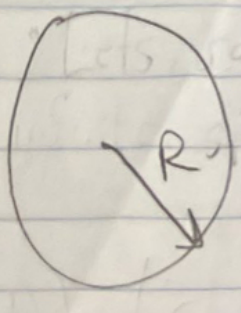
$$W_2 = \int_0^R \frac{Q_{enc}^2 r^4}{8\pi R^6 \varepsilon_0}\, dr = \frac{Q_{enc} r^5}{40\pi R^6 \varepsilon_0}\Big|_0^R$$

$$= \frac{Q_{enc}^2}{40\pi R \varepsilon_0}$$

$$W = \frac{5Q_{enc}^2}{40\pi\varepsilon_0 R} + \frac{Q_{enc}^2}{40\pi R\varepsilon_0} = \frac{6Q_{enc}^2}{40\pi\varepsilon_0 R}$$

$$\boxed{W = \frac{6Q_{enc}^2}{40\pi\varepsilon_0 R}}$$

- $W = \frac{1}{2}\int_V \rho(\vec{r})\,\phi(\vec{r})\,dV$



- $\rho(R) = \frac{3Q_{enc}}{4\pi R^3}$

$r \leq R$ region:

$$\frac{Q_{enc} r^2}{8\pi\varepsilon_0 R^3} + C$$

$r \geq R$:

$$-\frac{Q_{enc}}{4\pi\varepsilon_0 r}$$

$r = R$:

$$\frac{Q_{enc}}{8\pi\varepsilon_0 R} + C = -\frac{Q_{enc}}{4\pi\varepsilon_0 R}$$

$$-C = \frac{2Q_{enc}}{8\pi\varepsilon_0 R} + \frac{Q_{enc}}{8\pi\varepsilon_0 R}$$

$$C = -\frac{3Q_{enc}}{8\pi\varepsilon_0 R}$$

- $\phi(r) = \frac{Q_{enc} r^2}{8\pi\varepsilon_0 R^3} + \frac{3Q_{enc}}{8\pi\varepsilon_0 R}$
- $\rho(R) = \frac{3Q_{enc}}{4\pi R^3}$

$$W = \frac{1}{2}\left(\frac{3Q_{enc}}{4\pi R^3}\right)\left(\frac{Q_{enc}}{8\pi\varepsilon_0 R^3}\right)(4\pi)\int_0^R r^4\,dr + \frac{1}{2}\left(\frac{3Q_{enc}}{4\pi R^3}\right)\left(\frac{3Q_{enc}}{8\pi\varepsilon_0 R}\right)(4\pi)\int_0^R r^2\,dr$$

$$W = \frac{3Q_{enc}^2}{16\pi\varepsilon_0 R^6}\int_0^R r^4\,dr - \frac{9Q_{enc}^2}{16\pi\varepsilon_0 R^4}\int_0^R r^2\,dr$$

- $\int_0^R r^4\,dr = \frac{r^5}{5}\Big|_0^R = \frac{R^5}{5}$
- $\int_0^R r^2\,dr = \frac{R^3}{3}$

$$W = \frac{3Q_{enc}^2}{16\pi\varepsilon_0 R^6}\left(\frac{R^5}{5}\right) - \frac{9Q_{enc}^2}{16\pi\varepsilon_0 R^4}\left(\frac{R^3}{3}\right)$$

$$W = \frac{3Q_{enc}^2}{80\pi\varepsilon_0 R} + \frac{15Q_{enc}^2}{80\pi\varepsilon_0 R} = \frac{-12Q_{enc}^2}{80\pi\varepsilon_0 R}$$

$$W = \frac{-6Q_{enc}^2}{40\pi\varepsilon_0 R}$$

Both methods give the same magnitude of the stored energy, the negative sign occured because of choices of integrating, where I placed my C constants

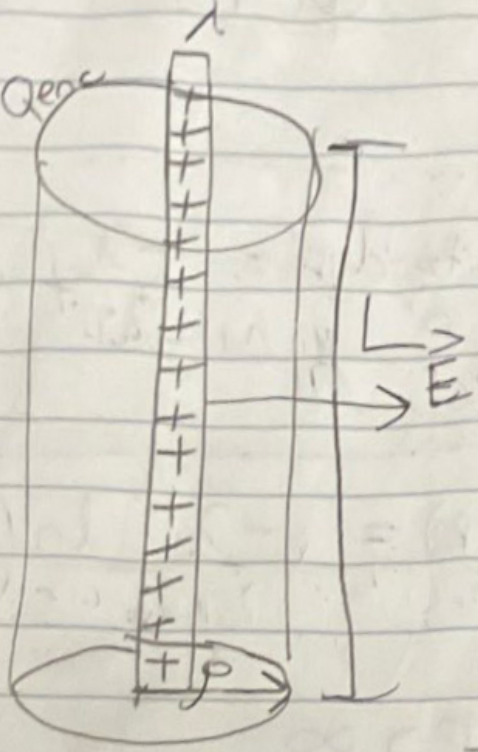
d.) • $W = \frac{6Q_{enc}^2}{40\pi\varepsilon_0 R}$

• $R \to 0$

$W = \infty$, so it's infinite energy at that point

Prob3.)

a.)



objective:

- a.) electric field of infinite Line charge.

coordinates (ρ, ϕ, z)

- Symmetry:
  - $\vec{E} = E_\rho \hat{\rho}$
  - $d\vec{l} = \rho d\phi dz \hat{\rho}$
  - $0 \leq \rho' \leq \rho$
  - $0 \leq \phi \leq 2\pi$
  - $0 \leq z \leq L$

- $\lambda L = Q_{enc}$

- $\int_A \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$

- $\rho E_\rho \int_0^{2\pi} \int_0^L d\phi dz = E_\rho \rho 2\pi L = \frac{Q_{enc}}{\varepsilon_0}$

- $\vec{E} = \frac{Q_{enc} \hat{\rho}}{2\pi\rho L \varepsilon_0} = \frac{\lambda L \hat{\rho}}{2\pi\rho L \varepsilon_0} = \frac{\lambda \hat{\rho}}{2\pi\rho\varepsilon_0}$

b.) $d\vec{l} = d\rho\hat{\rho}$

- $-\int_A^B \vec{E}\cdot d\vec{l} = \phi(\vec{r})$

- $\frac{-\lambda}{2\pi\varepsilon_0}\int_\infty^\rho \frac{1}{\rho'}\,d\rho' = \frac{-\lambda}{2\pi\varepsilon_0}\ln(\rho) = \phi(\rho)$

- $\phi(\rho) = \frac{-\lambda}{2\pi\varepsilon_0}\ln(\rho)$

- $\rho \to \infty$ the potential is undefined at infinity so it won't disappear.